① srcML of lines 14–15 and 18–20

```
<for pos:start="14:5">g for <control><expr><name>qubit</name></expr><range>in <expr><call>
    ↪ <name>range</name><argument_list>(<argument><expr><name>n b</name><operator>-</operator>
    ↪ <literal type="number">1</literal></expr></argument>)</argument_list></call></expr>
    ↪ </range></control><block>:<block_content>
        <expr_stmt><expr><call><name><name>qc</name><operator>.</operator><name>cx c</name>
            ↪ </name><argument_list>(<argument><expr><name>qubit d</name></expr></argument>,
            ↪ <argument><expr><name>qubit</name><operator>+</operator><literal type="number">1
            ↪ </literal></expr></argument>)</argument_list></call></expr></expr_stmt>
</block_content></block></for>

<expr_stmt pos:start="18:1">g <expr><call><name><name>qc</name><operator>.</operator><name>
    ↪ measure c</name><argument_list>(<argument><expr><name>n d</name></expr></argument>,
    ↪ <argument><expr><name>n e</name></expr></argument>)</argument_list></call></expr>
    ↪ </expr_stmt>
<with>f with <expr><call><name><name>qc</name><operator>.</operator><name>if_test f</name>
    ↪ </name><argument_list>(<argument><expr><tuple>(<expr><name><name>qc c</name><operator>.
    ↪ </operator><name>clbits</name><index>[<expr><name>n</name></expr>]</index></name></expr>,
    ↪ <expr><literal type="number">1</literal></expr>)</tuple></expr></argument>)
    ↪ </argument_list></call></expr><block>:<block_content>
    <expr_stmt><expr><call><name><name>qc</name><operator>.</operator><name>x c</name>
        ↪ </name><argument_list>(<argument><expr><literal type="number">0 d</literal></expr>
        ↪ </argument>)</argument_list></call></expr></expr_stmt>
</block_content></block></with>
```

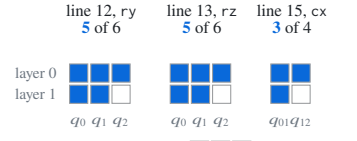
a `for` → unrolled **b** range → iteration count, from line 4 **c** gate **d** qubit argument, evaluated per iteration
e classical bit, written by the measurement and read by the condition **f** `if_test` → conditioned block **g** position → source line

② Trace

line	iteration	operation
12, 13	every	12 rotations
15	0, 0	cx q_0, q_1
15	0, 1	cx q_1, q_2
15	1, 0	cx q_0, q_1
15	1, 1	cx q_1, q_2
17	–	h q_3
18	–	measure $q_3 \rightarrow c_3$
20	–	x q_0 if c_3

An iteration lists the values of `layer` *and* `qubit`*. The measurements of line 21 are terminal and not shown.*

③ Slice of q_0



Filled squares are in the QDG slice of q_0, 16 of 19 operations. Every iteration runs the same lines, but in the second layer the rotations of q_2 and its `cx` *are not dependences of q_0, and a slice of lines keeps lines 12, 13 and 15 anyway.*